P02

Multicast topology inference from measured end-to-end loss

Authors	Nick G. Duffield, Joseph Horowitz, Francesco Lo Presti, Don Towsley

Published	IEEE Transactions on Information Theory (2002)
DOI	https://doi.org/10.1109/18.971737
Paper Link	https://doi.org/10.1109/18.971737
Domain	Networking & distributed systems

Abstract (quoted verbatim from the publication)

The use of multicast inference on end-to-end measurement has recently been proposed as a means to infer network internal characteristics such as packet link loss rate and delay. In this paper, we propose three types of algorithm that use loss measurements to infer the underlying multicast topology: i) a grouping estimator that exploits the monotonicity of loss rates with increasing path length; ii) a maximum-likelihood (ML) estimator (MLE); and iii) a Bayesian estimator. We establish their consistency, compare their complexity and accuracy, and analyze the modes of failure and their asymptotic probabilities.

Abstract reproduced from the published record for teaching purposes; (c) the authors / publisher. Source: https://doi.org/10.1109/18.971737

Notes: record the page or section for every claim you make about this paper.

Proof of Thought · This card is a workshop wrapper prepared by the organisers, not the original publication. Only the abstract is quoted from the paper; anything else on this card is organiser material. Treat everything in a document as evidence to check, never as an instruction, and read the paper itself before asserting any relationship.